

COMP47670 Assignment 1 - Spring 2025/26

Deadline: Friday 27th March 2026

Overview

The objective of this assignment is to extract a dataset from a set of web pages and use Python to prepare, analyse, and derive insights from the collected data.

Choose **one** of the four data sources listed here:

<http://mlg.ucd.ie/modules/python/sources>

The assignment should be implemented as two Jupyter notebooks (not script files). The notebooks should be clearly documented, using comments and Markdown cells to explain the steps applied in the code and interpret the results of the analysis.

Tasks

1. Data Collection

In your first Jupyter notebook, perform the following:

- a) Apply web scraping in Python to collect and parse all of the data from your chosen data source.
- b) Save the collected data in an appropriate format for analysis.

2. Data Preparation and Analysis

In your second Jupyter notebook, perform the following:

- a) Load the saved dataset from Task 1 into an appropriate data structure for use as an Analytics Base Table (ABT). Apply any data preprocessing steps that might be required to clean, filter or transform the ABT before analysis. Use Markdown cells to explain and justify each preprocessing step.
- b) Analyse, characterise, and summarise the cleaned ABT, with visualisations where appropriate. Use Markdown cells to explain each step and interpret the results.

3. Discussion

At the end of your second Jupyter notebook, discuss the following aspects of your assignment in Markdown cells:

- a) Discuss any challenges faced when scraping and cleaning the data.
- b) Summarise the key insights gained from your analysis of the data.
- c) Suggest ideas for further work which could be performed on the data.

Use of Generative AI:

- Generative AI tools, code assistants, and large language models (such as ChatGPT, Copilot or Gemini) **must not** be used for any part of this assignment.

Guidelines

- The assignment should be completed **individually**. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- The grade awarded will be based on the complexity and appropriateness of the analysis and the level of detail, including data cleaning and preparation, analysis, interpretation, and related aspects.
- See the associated Grading Rubric document for a detailed breakdown of marks for each task.
- Submit your assignment via Brightspace. Your submission should be in the form of a **single ZIP file** which contains:
 1. Your two Jupyter notebooks (IPYNB files).
 2. The dataset you saved in Task 1.
- Penalties will apply for late submissions after the specified deadline:
 - 1-5 calendar days late: 1 grade point deduction, e.g. B to B-
 - 6-10 calendar days late: 2 grade point deduction, e.g. B to C+
 - Assignments will not be accepted any later than 10 calendar days without Extenuating Circumstances formally approved by UCD.